Haggaï Bonneau

Academic curriculum vitae · Paris, France

LinkedIn: linkedin.com/in/haggai-bonneau
Google Scholar: scholar.google.com/citations?user=iTKDGOEAAAAJ

RESEARCH INTERESTS

Non-equilibrium statistical physics · Stochastic processes · Interacting many-body systems · Electrolytes · Polymer physics · Random graphs and complex networks

RESEARCH EXPERIENCE

Career break for family caregiving 2025-2026

Postdoctoral researcher 2024-2025

Tel Aviv University & IPhT Saclay

- Developed stochastic field-theory descriptions of concentrated electrolytes and derived analytical predictions for their frequency-dependent conductivity and correlation functions.
- Studied how electrostatic, steric, and hydrodynamic interactions affect the response of ionic systems to time-dependent electric fields.
- Explored coupled ordinary differential equations numerically in Python to develop physical intuition alongside the analytical calculations.

Doctoral researcher 2020-2023

Gulliver Laboratory, ESPCI Paris · Université PSL

- Studied transient response and correlations in driven electrolytes using stochastic density functional theory and fluctuating hydrodynamics.
- Investigated the connection between algebraic relaxation of electric current and the diffusive dynamics of non-equilibrium correlations.
- Used first-passage and random-walk statistics to test a proposed relaxation mechanism in polymer films, in collaboration with the Forrest group at the University of Waterloo.
- Analyzed discrete-element simulation results and contributed to a scaling description connecting microscopic grain motion to macroscopic granular rheology.

Research assistant 2016-2018

Racah Institute of Physics, Hebrew University of Jerusalem

- Studied random-graph structure using generating functions and cavity/message-passing methods, including shortest cycles and articulation points.
- Generated numerical graph ensembles and compared structural statistics with analytical predictions. This research continued during the MSc.

EDUCATION

PhD in Physics 2023

Université PSL, prepared at ESPCI Paris · Gulliver Laboratory

Thesis: *Out of equilibrium dynamics in charged and soft matter systems*.
Advisors: Élie Raphaël and Vincent Démery.

MSc in Physics, with honours 2020

Hebrew University of Jerusalem · Academic Excellence Award

BSc in Physics, with honours 2018

Hebrew University of Jerusalem

TEACHING EXPERIENCE

Teaching assistant & laboratory instructor 2018-2021

ESPCI Paris & Hebrew University of Jerusalem

- **Event-driven molecular dynamics, ESPCI Paris:** taught the implementation of basic simulations in Python.
- **Electromagnetism, Hebrew University:** taught tutorials supporting the physics course.
- **Superconductivity laboratory, Hebrew University:** supervised practical experiments involving resistance measurements, Hall probes, and temperature monitoring.

PUBLICATIONS

H. Bonneau, V. Démery, E. Raphaël.
Stationary and transient correlations in driven electrolytes.
Journal of Statistical Mechanics: Theory and Experiment (2025), 033201.

H. Bonneau, Y. Avni, D. Andelman, H. Orland.
Frequency-dependent conductivity of concentrated electrolytes: A stochastic density functional theory.
Journal of Chemical Physics **161**, 244501 (2024).

H. Bonneau, V. Démery, E. Raphaël.
Temporal response of the conductivity of electrolytes.
Journal of Statistical Mechanics: Theory and Experiment (2023), 073205.

H. Bonneau, M. Arutkin, R. Chen, J. A. Forrest, E. Raphaël, T. Salez.
On the bridge hypothesis in the glass transition of freestanding polymer films.
European Physical Journal E **46**, 8 (2023).

D. Dumont, **H. Bonneau**, T. Salez, E. Raphaël, P. Damman.
Microscopic foundation of the μ(I) rheology for dense granular flows on inclined planes.
Physical Review Research **5**, 013089 (2023).

H. Bonneau, I. Tishby, O. Biham, E. Katzav, R. Kühn.
Fate of articulation points and bredges in percolation.
Physical Review E **103**, 042302 (2021).

H. Bonneau, O. Biham, R. Kühn, E. Katzav.
Statistical analysis of edges and bredges in configuration model networks.
Physical Review E **102**, 012314 (2020).

H. Bonneau, A. Hassid, O. Biham, R. Kühn, E. Katzav.
Distribution of shortest cycle lengths in random networks.
Physical Review E **96**, 062307 (2017).